

C. Now only consider the **Means of both tests** as the outcome. Dr. W claims these means are **not normally distributed**. Assume Dr. W's claim is true and do a two sided test of the null hypothesis H_0 : that for a given person that the median of the two test means is 4 units higher for the new test than for the standard test against the alternative that the median difference is something else. Use a type-1 error of 0.05.

3. Questions (a)-(c) involve analysis of data on cigarette consumption from 50 states of the USA. The variable *perfem* is the percentage of persons in a state who are female. A simple linear regression model was fit to predict the weekly cigarette consumption (per person) using the percentage of females and this partial output was obtained:

Coefficients:

	Value	Std. Error	t-Value	p value
(Intercept)	-93.4260	207.8126	-0.4496	0.6550
<i>perfem</i>	4.2191	4.0777		0.3059

Residual standard error: 32.05 on 49 degrees of freedom

Multiple R-Squared: 0.02138

- Calculate the t-statistic for *perfem*. What (if any) is the unit of the t-statistic? Comment on the effect of *perfem*.
- Suppose we replaced the percentage of females by the percentage of persons in the state who are male in the above regression model. Give the regression coefficients values for the new model. Would the standard errors for; i) the slope change and ii) the intercept change why or why not?
- When we look at the ANOVA table for a regression model, it is possible to have a p value of exactly zero in an F-statistic for a regression? Explain clearly why (how) this could happen or why it is impossible.

4. The following data were obtained in a comparison of two different methods for determining dissolved oxygen concentration (in milligrams per liter) on the same samples.

Oxygen Concentration Data

Sample #	1	2	3	4	5	6
Method A	2.62	2.65	2.79	2.83	2.91	3.57
Method B	2.73	2.80	2.87	2.95	2.99	3.67

- Perform an appropriate test at the 0.05 level of the null hypothesis that the two methods give the same dissolved oxygen concentration. Formulate this hypothesis and give details of your computations. Give all assumptions needed.
- Give a 95 percent confidence interval for the mean difference of the two methods.
- Suppose now that different samples were tested for Method A and B. Perform an appropriate test at the 0.05 level. Give all assumptions needed. Comment on the difference between the test here and the one in part (a). What has caused the difference?